

Project 6.5 Model a Miniature Train

Introduction

Have you ever ridden on a train or owned a train set? The parts that make up the engine car on a train can vary depending on the make and model; however, all train engine cars have parts that are similar.

Interpreting dimensioned drawings is an important engineering skill. Using drawings to create a computer model of a part is also important. You learned earlier in this course that a sketch is the documentation foundation for related technical work. Communicating this information effectively allows a group of people to function as a design team.

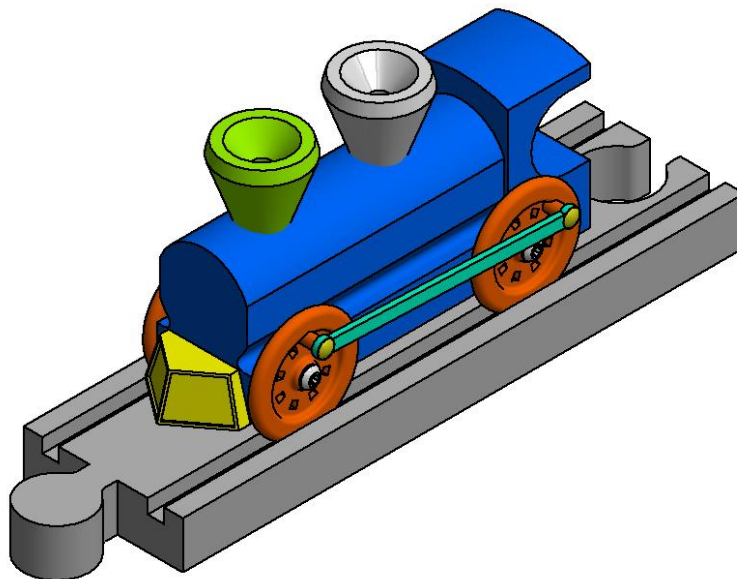
In this project you will further develop your modeling skills and your ability to use a computer as an efficient communication tool. The skills that you learned earlier in this course will be systematically applied to model the eight remaining parts needed for the Miniature Train Assembly. The parts with the dimensions are listed below.

Caution: These parts are different from what you might find on the internet!

Procedure

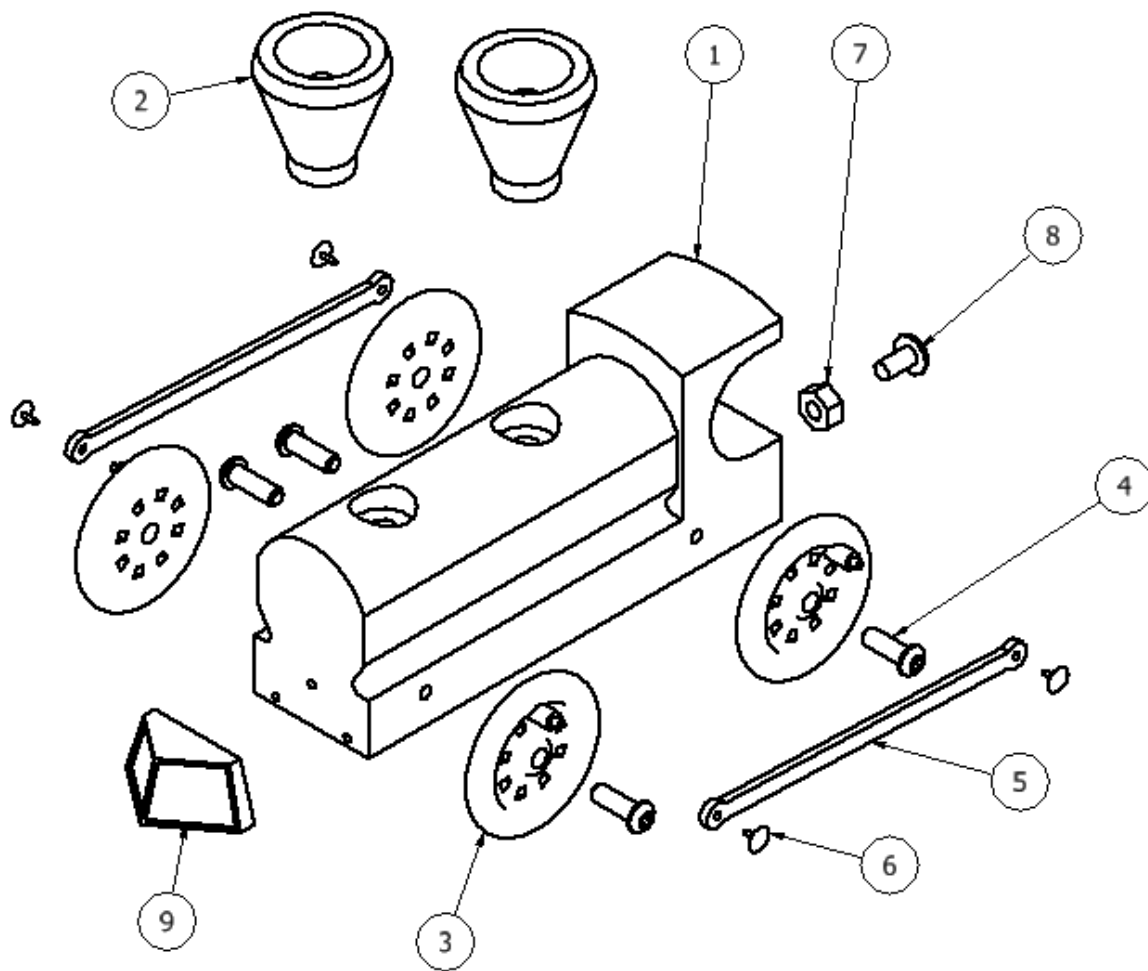
1. Model and assemble the parts shown using the drawings provided.
2. Create a set of working drawings to document the train parts and assembly.
3. Follow the rubric for maximum success. **Compile documents in the order shown on the rubric**

The assembled train looks like this.



Train Parts List

Item	Quantity	Name	Material
1	1	Train Body	PLA Plastic
2	1	Stack	PLA Plastic
5	4	Wheel	PLA Plastic
6	4	Axle Peg	PLA Plastic
7	2	Linkage Arm	PLA Plastic
8	4	Linkage Peg	PLA Plastic
3	1	Hitch Magnet	PLA Plastic
4	1	Hitch Peg	PLA Plastic
9	1	Cow Catcher	PLA Plastic
10	1	Track	PLA Plastic



Train Tolerances

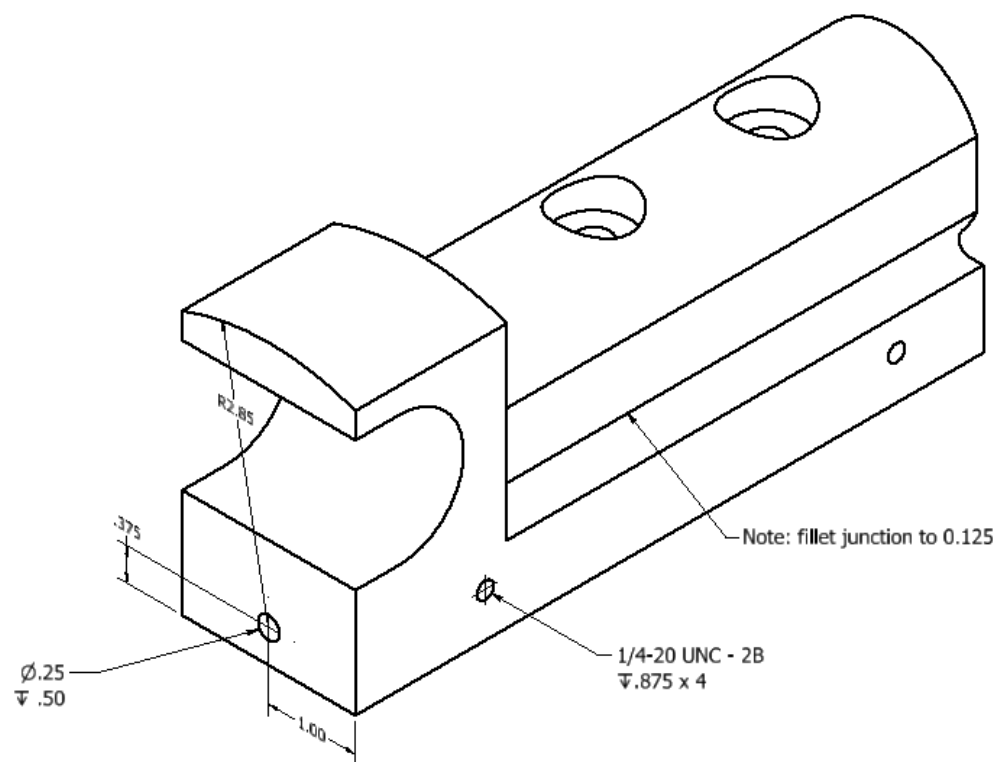
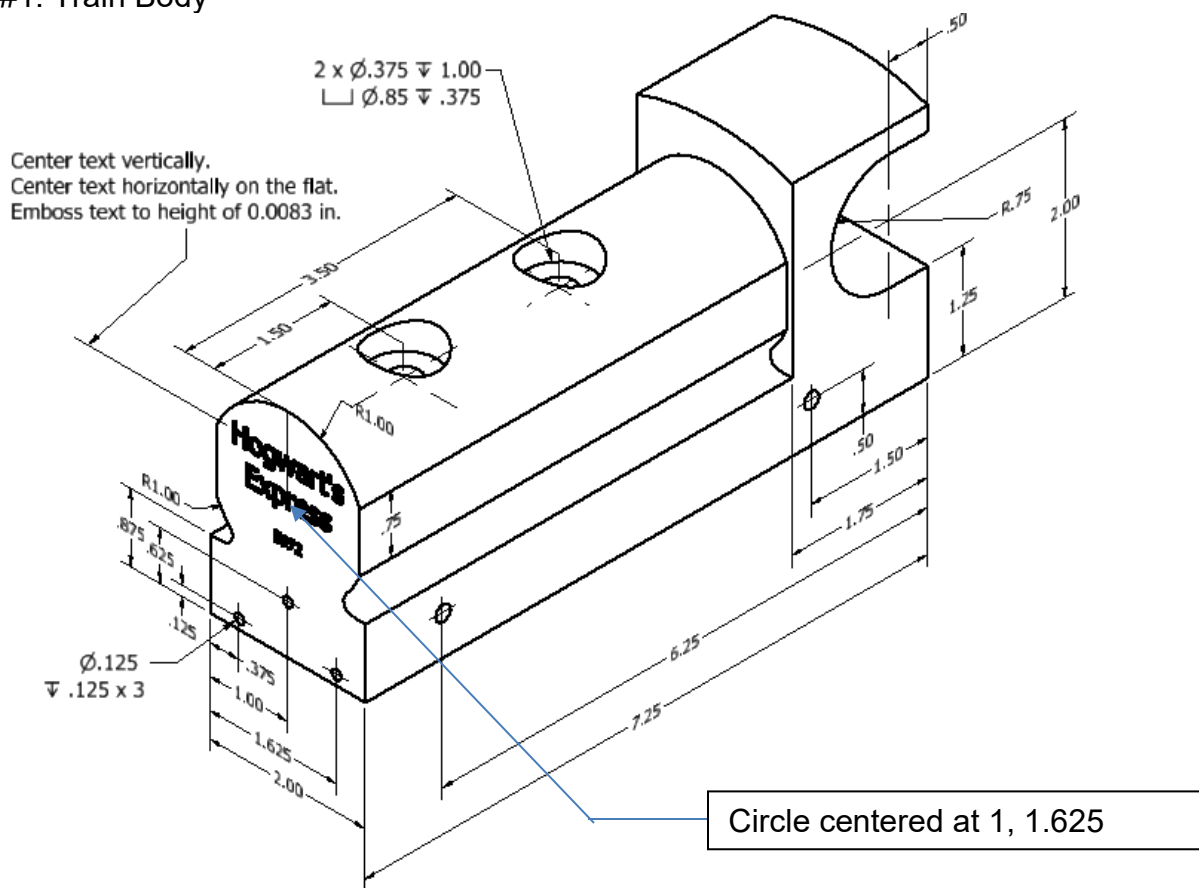
All parts have the following tolerances:

X.X = +/- .020

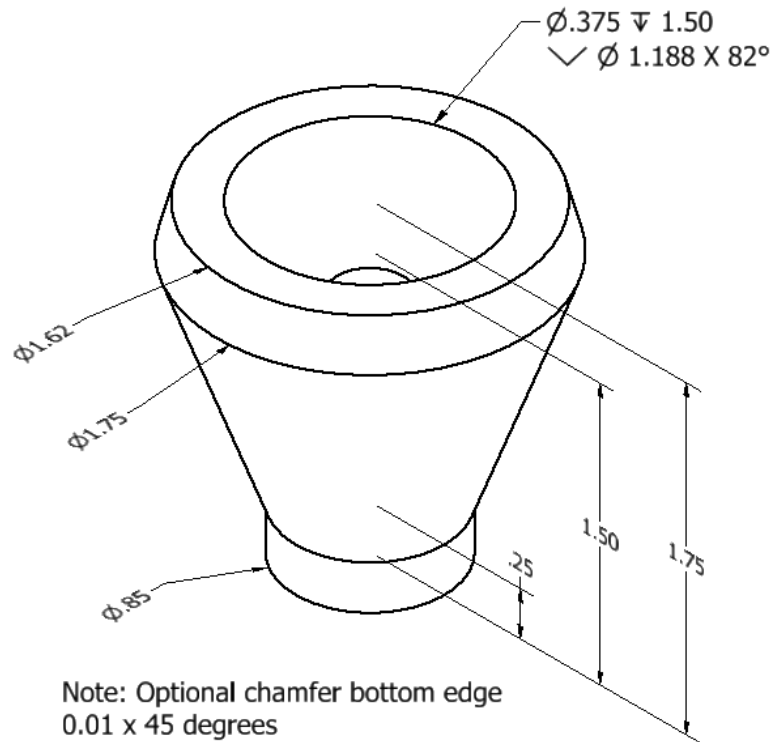
X.XX = +/- .010

X.XXX = +/- .005

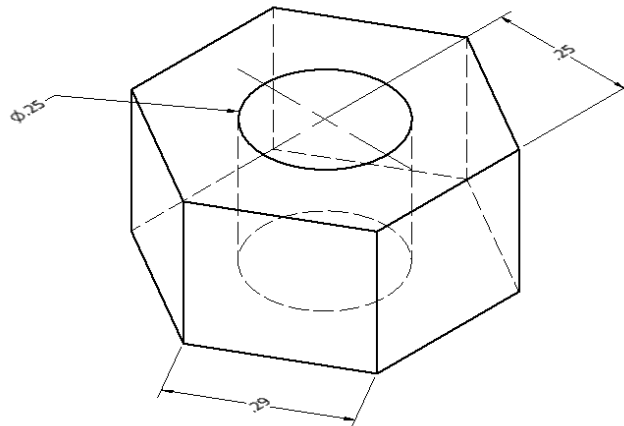
Part #1: Train Body



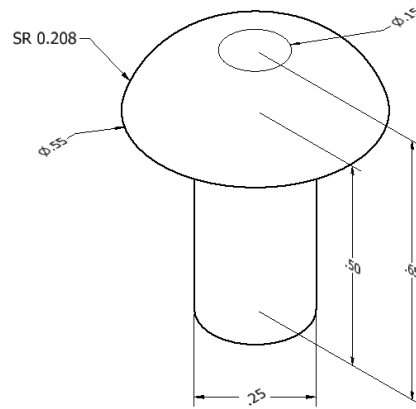
Part #2: Stack



Part #3: Hitch Magnet (thickness is 0.25")

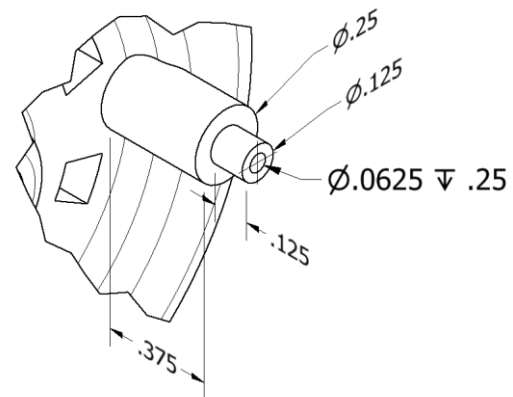
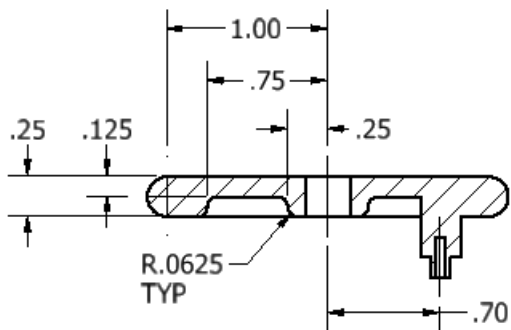
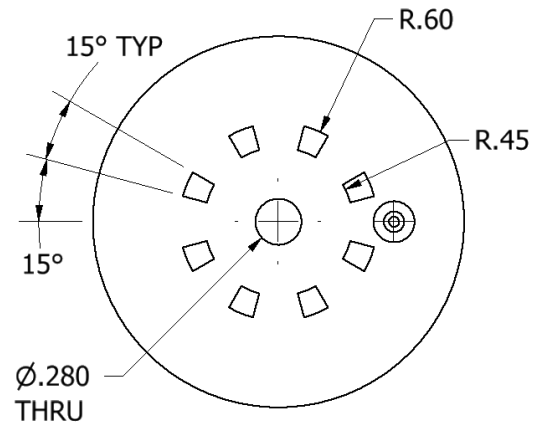
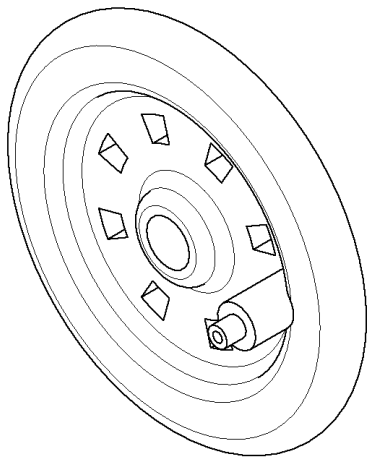


Part #4: Hitch Peg

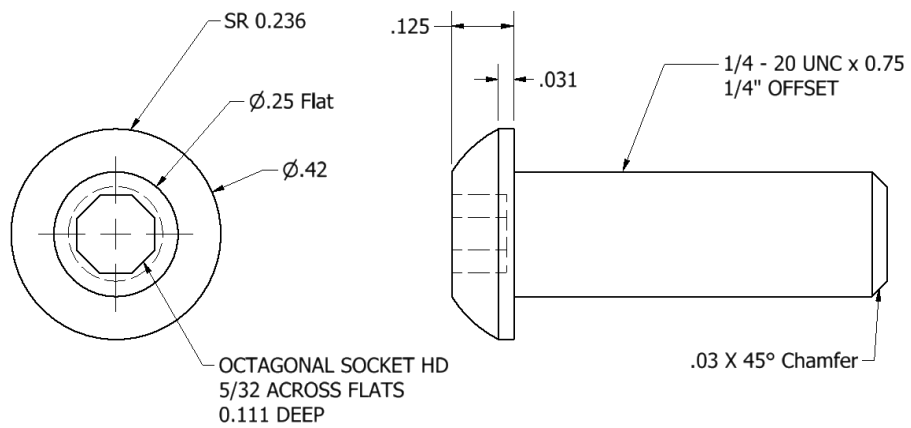


Note: Optional chamfer bottom edge .01 x 45 deg

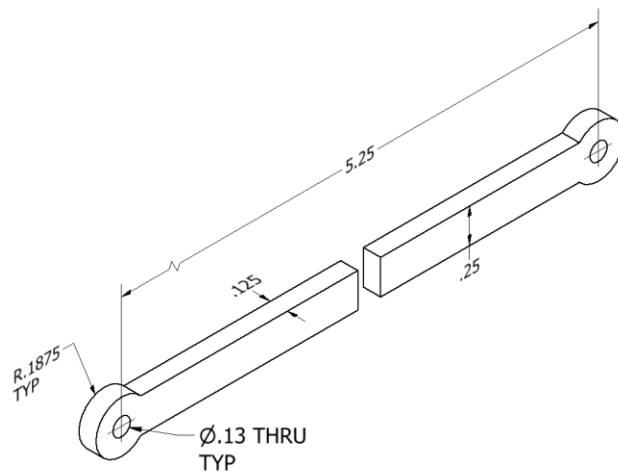
Part #5: Wheel



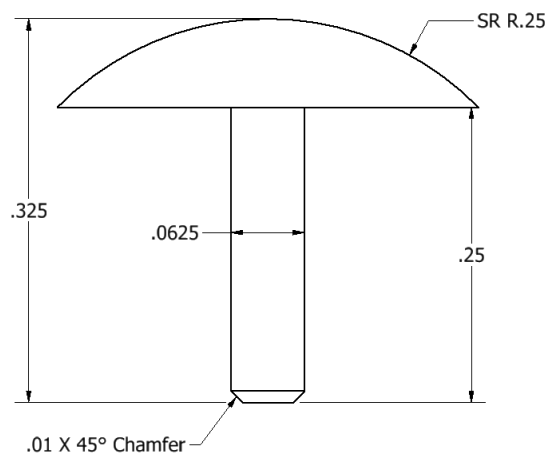
Part #6: Axle Peg



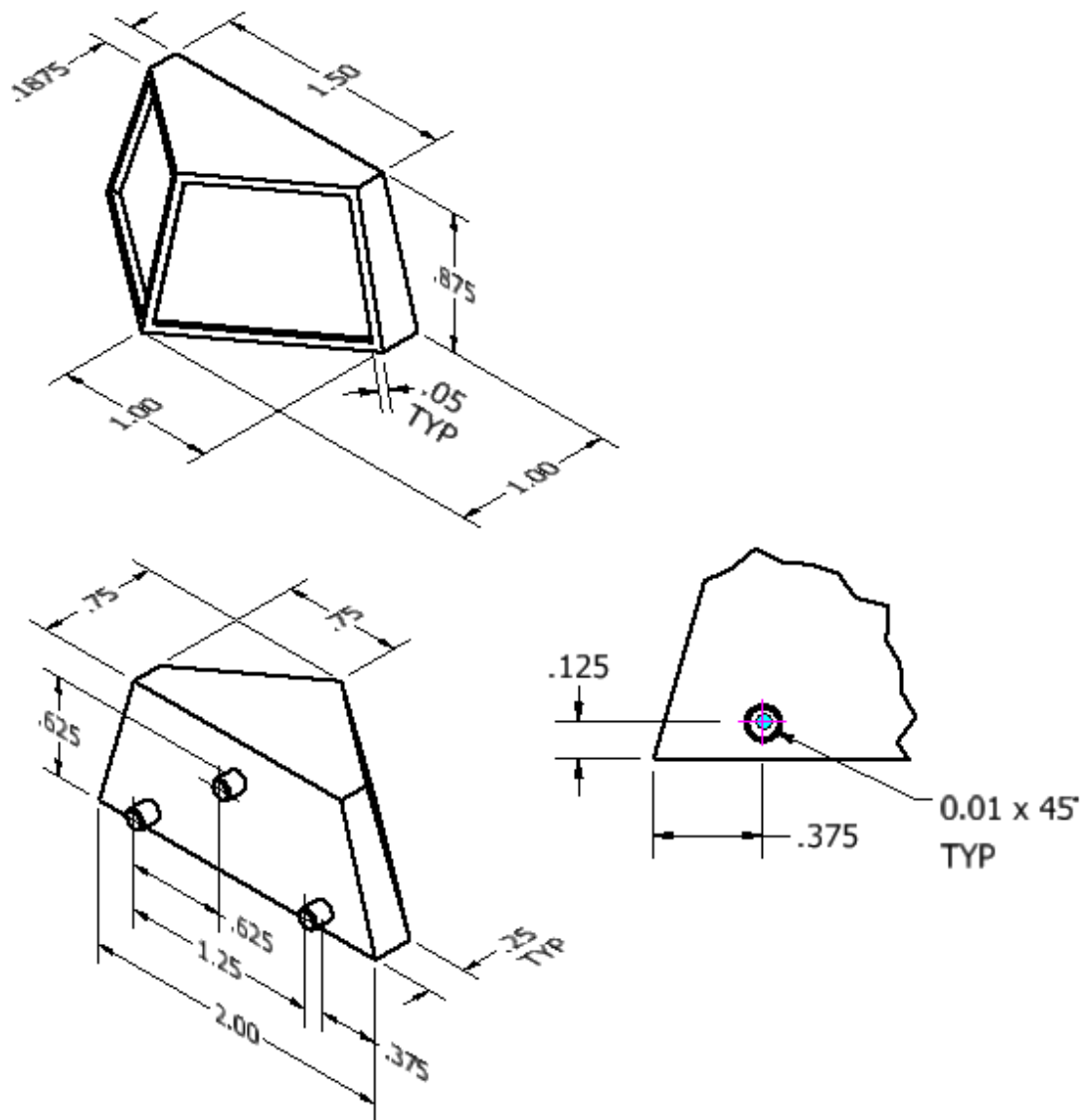
Part #7: Linkage Arm



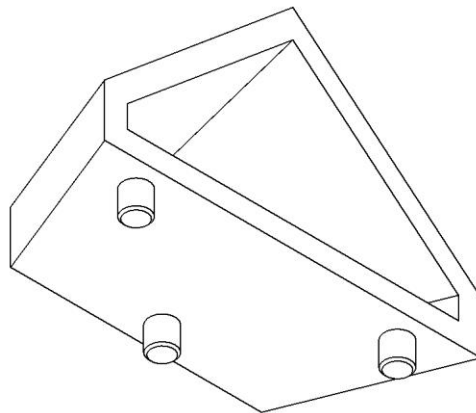
Part #8: Linkage Peg



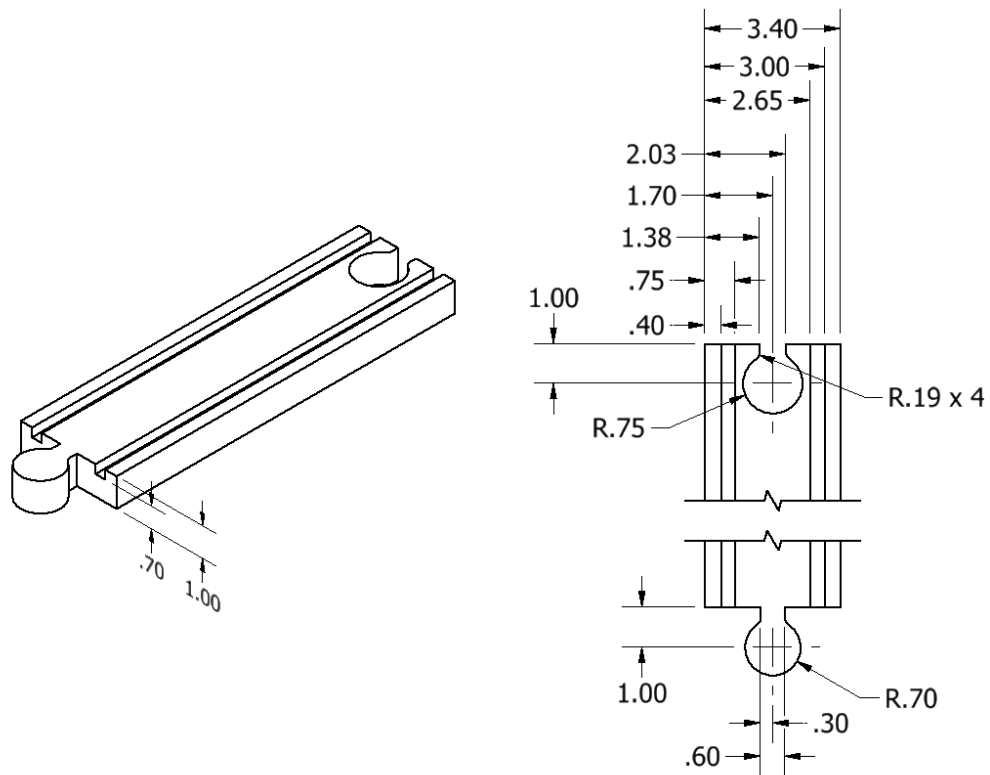
Part #9: Cow Catcher



Note: part shelled to 0.1 TYP



Straight Track



Assembled Train

